

Fine-Tuning BERT for SMS Spam Detection

A Transformer-based approach to safer digital communication

1. Task Selection and Domain Justification

This project fine-tunes a pretrained BERT model to classify SMS messages as ham or spam. The task matters because spam messages often contain scams, phishing attempts, account warnings, and other fraud cues that can harm users and organizations. Improved spam detection gives communication platforms a way to flag high-risk messages earlier, reduce manual review volume, and protect users from clicking suspicious links or sharing private information. Compared with recurrent models, Transformers are better suited for this task because self-attention lets the model evaluate all tokens in a message together, and BERT uses bidirectional context so each word is interpreted using both earlier and later words. The domain is challenging because SMS messages are short, informal, noisy, and sometimes intentionally disguised with unusual spelling or urgent language.

2. Dataset and Preprocessing

The project uses the Hugging Face [ucirvine/sms_spam](#) dataset, which contains 5,574 labeled SMS messages. Labels are binary: ham and spam. The data was split using stratified sampling so that the class distribution remained consistent across train, validation, and test sets.

Split	Ham	Spam	Total
Train	3,861	598	4,459
Validation	483	74	557
Test	483	75	558

Text was tokenized with the BERT tokenizer using attention masks and a maximum sequence length of 128 tokens. Because spam messages are less common than ham messages, weighted cross-entropy loss was used to reduce the impact of class imbalance.

3. Model Architecture and Training Setup

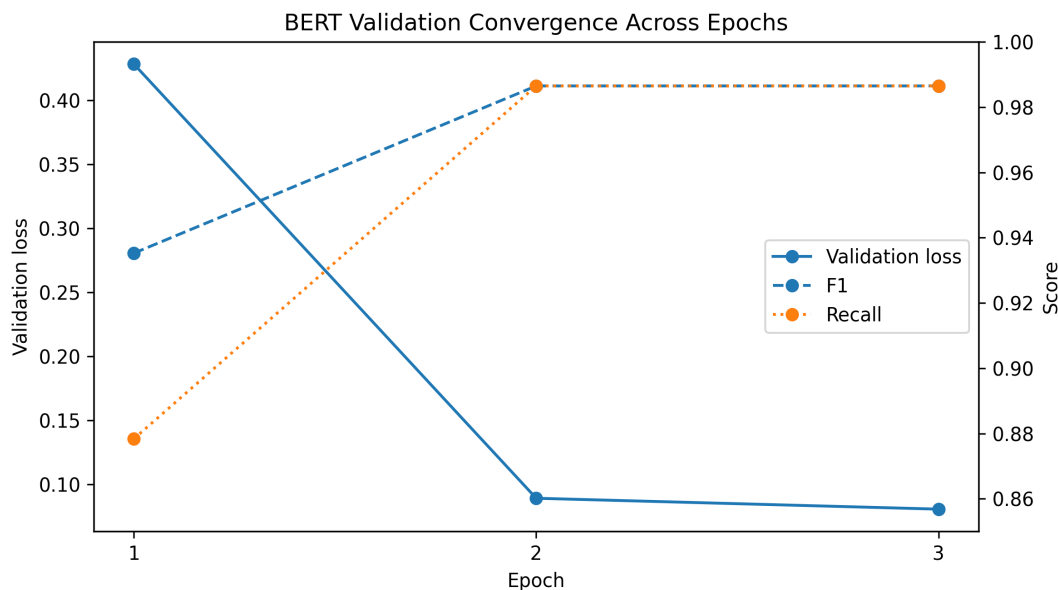
The model is [google-bert/bert-base-uncased](#) with a sequence classification head added for the two labels. The BERT encoder uses stacked Transformer layers, multi-head self-attention, residual connections, and bidirectional contextual encoding. The classification head converts the contextual representation into a ham or spam prediction.

Item	Setting
Base model	google-bert/bert-base-uncased
Task type	Binary sequence classification
Max sequence length	128 tokens
Epochs	3
Optimizer / schedule	AdamW with linear learning-rate scheduling
Learning rate	2e-5
Batch size	16
Stability controls	Gradient clipping and weighted loss

4. Training Behavior and Validation Results

Most learning happened between Epoch 1 and Epoch 2. Validation loss dropped from 0.4283 to 0.0892, while F1 increased from 0.9353 to 0.9865. Epoch 3 preserved the same classification metrics and slightly lowered validation loss, suggesting convergence by Epoch 2.

Epoch	Training Loss	Validation Loss	Accuracy	Precision	Recall	F1
1	0.014724	0.428344	0.983842	1.000000	0.878378	0.935252
2	0.000030	0.089155	0.996409	0.986486	0.986486	0.986486
3	0.033790	0.080646	0.996409	0.986486	0.986486	0.986486



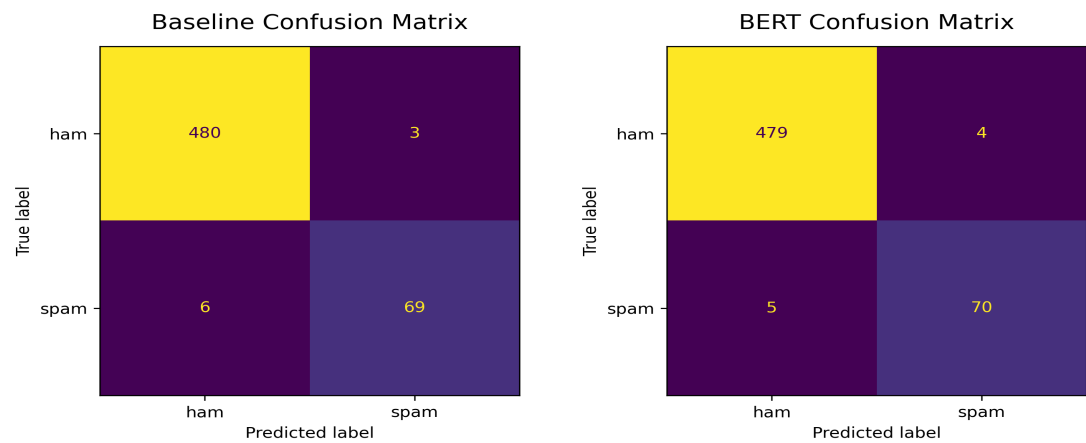
5. Test Set Model Comparison

The baseline model is TF-IDF with logistic regression. On the test set, both models reached 98.39 percent accuracy. BERT slightly improved spam recall and F1 while producing a much higher AUC, which is useful because thresholds can be tuned for different risk policies.

Model	Accuracy	Precision	Recall	F1	AUC
TF-IDF + Logistic Regression	0.9839	0.9583	0.9200	0.9388	0.9759
Fine-tuned BERT	0.9839	0.9459	0.9333	0.9396	0.9977

6. Confusion Matrix Interpretation

The baseline correctly classified 480 ham and 69 spam messages, with 3 false positives and 6 false negatives. BERT correctly classified 479 ham and 70 spam messages, with 4 false positives and 5 false negatives. The key operational improvement is one fewer missed spam message.



7. Technical to Business Impact

Technical result: BERT delivered high accuracy, stronger spam recall than the baseline, and the highest AUC. **Business meaning:** higher recall means fewer harmful messages slip through, while high AUC makes the model easier to threshold for different risk policies. **Workflow improvement:** low-risk ham can pass automatically, high-confidence spam can be blocked or quarantined, and uncertain messages can be routed to human review. This improves efficiency while keeping oversight where risk is highest.

Technical Metric	Operational Meaning
Accuracy: 98.39%	Most messages are routed correctly.
Spam recall: 93.33%	Most spam is caught before reaching users.
AUC: 0.9977	Thresholds can be tuned for business risk.
False negatives: 5	Remaining missed spam should be monitored and reviewed.

8. Limitations and Future Work

The model may fail on new scam templates, code-switching, misspellings, extremely short messages, or messages that rely on external context such as URLs or sender history. Future improvements include adding multilingual and adversarial spam examples, calibrating the spam decision threshold, using DistilBERT for faster deployment, monitoring model drift, and combining message text with metadata such as sender reputation and link risk signals.

References: Hugging Face dataset *ucirvine/sms_spam*; Hugging Face model *google-bert/bert-base-uncased*; Devlin et al., 2019, BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding.